ECI v7.6 amendment: NPP20 lepton sector and G1.12.B SU(5) modular $f^{ij}(\tau = i)$ quark sector, with Karam–Palatini Cassini reference, King–Wang dS-trap mechanism, CSD$(1+\sqrt{6})$ leptogenesis, and Phase 3 $N = p^2$ interlinking + real-data ECI posterior

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Abstract

The v7.4 amendment of the Extended Cosmological Index (ECI) research programme [1] relaxed the strict S-fixed point axiom $H_5 \to H_5'$ (CM-anchored attractor) and replaced the Damerell–Coates–Sinnott bridge $H_7 \to H_7'$ by a two-part bridge (integer-point Damerell ladder + BDP p-adic complement). The present v7.5 amendment records six concrete extensions, one retraction, and one structural reframing.

Retraction (A46): the v7.4 "LYD20 unified scaffold with τ pinned at $\tau_S = i$" graft is empirically REFUTED. A 19-observable joint quark+lepton refit (PDG 2024) yields χ^2 penalty $\approx 99{,}061\times$ versus the LYD20 published best fit at $\tau_{\mathrm{LYD20}} = -0.21 + 1.52\,i$ – five orders of magnitude beyond the $\sim 1.8\times$ tolerance estimated by the earlier A42 scoping. v7.5 therefore adopts Option C: NPP20 lepton-only at $\tau_S = i$ (4–6 free parameters for 9 observables) plus the G1.12.B SU(5) $5_H + 45_H$ modular $f^{ij}(\tau = i)$ campaign for the quark sector, DROPPING the unified scaffold.

Grafts: (1) a stand-alone Cassini-wall section grafting the Karam–Sánchez López–Terente Díaz analytical PPN reference (KSTD 2026, `arXiv:2604.16226`), with v7.5-P Palatini sub-branch registered as a conditional 2027-decision direction; (2) the King–Wang multi-modulus dS-trap (KW 2023, `arXiv:2310.10369`) supplying a string-cosmological mechanism for $\tau \to i, \omega$ vacua (closes A14's "WHY $\tau = i$?" gap); (3) the dMVP26 canonical-Kähler quark hierarchy at $\tau = i$ (A48 PARTIAL graft) with $J^q_{CP} = 0$ cost flagged; (4) leptogenesis from the CSD$(1+\sqrt{6})$ Littlest Modular Seesaw at $\tau_S = i$ (A55) yielding the structural signature $(n-1)^2 = 6$ in ϵ_1 scaling and Planck-compatible $Y_B^{\mathrm{calib}} = 0.872 \times 10^{-10}$.

Structural reframings: (5) the A42 "two-τ is $60\times$ worse" claim is corrected per A49 autopsy of Du–Wang `arXiv:2209.08796` – the comparison was a category error; two-τ is GENERICALLY HELPFUL empirically. v7.5 keeps single-τ on STRUCTURAL grounds (Du–Wang's GUT matter-unification orbifold-BC reduction), not numerical χ^2. (6) The A54 BC$\times$CM at $\beta = 2\pi$ chain (Connes–Marcolli–Ramachandran 2005, `arXiv:math/0501424`) is honestly downgraded: it does NOT bypass H_6 (χ_4 nebentypus restriction) and joins the Galois-descent chain (P-NT) as a SECOND H_6-co-dependent partial derivation, not two independent analytical derivations.

We collect seven experimental disambiguators 2027–2045 (NA62, JUNO, DUNE, KamLAND-Zen, CMB-S4, Belle II, Hyper-K) in Section 11 and list open questions in Section 13. The arithmetic-modular core ($f_{4.5.b.a}$, $\mathbb{Q}(i)$, Damerell ladder, CSD$(1+\sqrt{6})$, G1.12.B M1–M6) is preserved across the v7.4 $\to$ v7.5 transition.

Honest negative result (A72, post-submission): A null test (A72, 2026-05-06) finds that the $K = \mathbb{Q}(i)$ Damerell-ladder numerology is statistically indistinguishable from random rational ladders on the full PMNS+lepton+quark observable space ($Q(i) = 358$ vs null $= 366 \pm 25$, $\sigma = -0.33$, $P = 0.63$, strict cut). The Damerell ladder survives as an *arithmetic anchor* for $f_{4.5.b.a}$ (LMFDB 4.5.b.a, Hecke H_1, χ_4 nebentypus) but does NOT constitute a predictive structural input for low-energy fermion observables beyond what random rationals achieve. Sections 8 and 12 document this downgrade.

v7.6 deltas (Phase 3 of 2026-05-06, 23+ sub-agent atterrissages): (i) $N = p^2$ *structural interlinking* (M22 META-finding, §3.4): three previously-independent observations on $f_{4.5.b.a}$ — the critical-slope identity $a_2 = -2^{(k-1)/2}$, the functional-equation pair-sum $\alpha_2+\alpha_3 = 2^{-3}$, and the β-renormalised monotone v_2-pattern $\{-3,-2,0,+1\}$ — are unified by the unique fact that 4.5.b.a is the only CM-by-$\mathbb{Q}(i)$ weight-5 newform with level $N = p^2 = 4$ in the small-level regime; the critical-slope condition is reframed as classical (Shimura 1971, Ribet 1977) per audit M20. (ii) *First real-data ECI posterior* (§4): 4-chain NUTS on DESI DR2 + Pantheon+ + Planck 2018 yields $H_0 = 68.51 \pm 0.14$ km/s/Mpc (*Planck-like*) and a ξ_χ posterior *rail-saturated* at the Cassini boundary, diagnosed as a *real data signal* (M24) consistent with A73 RG-running. (iii) *Conjecture M13.1* (§10): a Pollack-type rescue conjecture for the 2-adic $L_2^{\pm}(f)$ of $f_{4.5.b.a}$ via the Kriz 2021 Hodge-filtration framework (Princeton AMS-212), bypassing the $X(X+4)$ Frobenius zero-root degeneracy. Targeted as math.NT paper-2 (*Algebra & Number Theory* tier). (iv) $B_{\text{Higgs}} = 1.077$ *modular-geometry insight* (M23): a κ_u-INDEPENDENT prediction $B_{\text{Higgs}} = f^{uu}(\tau^*)/f^{uc}(\tau^*) = 0.895$ at the W1 attractor, lifted to $B \sim 2.06$ by gauge X, Y exchange. (v) Honest tensions audit, SUSY decision (NO-GO), and 6+ papers locked SUBMISSION-READY with M23 prose fixes documented.

1 Introduction: from v7.4 to v7.5

The v7.4 amendment [1] closed three open questions of the v7.3 axiomatic restart of ECI [13]: it relaxed the strict S-fixed point H_5 to a CM-anchored attractor H_5' ($|\tau - i| \lesssim 0.2$, supported by a 30×30 χ^2 scan W1 yielding $\tau^* = -0.1897 + 1.0034\,i$ at $\chi^2/\text{dof} = 1.05$); replaced the half-integer Damerell–Coates–Sinnott bridge H_7 – refuted by the G1.15 audit on parity grounds for an odd-weight CM newform – by an integer-point Damerell ladder H_7' ($\alpha_m \in \{1/10, 1/12, 1/24, 1/60\}$ for $m \in \{1,2,3,4\}$) with parameter-free Cardy hits $\rho_{\text{Cardy}}(c = 1) = 1/12$ (free boson) and $\rho_{\text{Cardy}}(c = 1/2) = 1/24$ (Ising); and recorded six wave 4–5 follow-ups (A14, A16, A17, A22, A24, A26).

The present v7.5 amendment records the post wave 4–5 development of 2026-05-05 (evening and night), comprising one retraction (A46), four grafts (A47, A48, A50, A55), and two structural reframings (A49, A54). Section 2 grafts the Karam–Sánchez López–Terente Díaz analytical PPN reference for the Cassini wall (KSTD 2026, `arXiv:2604.16226`) and registers a Palatini sub-branch v7.5-P as a conditional 2027-decision direction. Section 5 collects the v7.5 modular flavour body in Option-C form (NPP20 lepton + G1.12.B quark), with subsections covering: G1.12.B SU(5) M1–M6 closure including the M5 partial-width computation (Sec. 5.2); the CSD$(1+\sqrt{6})$ Littlest Modular Seesaw at $\tau_S = i$ (Sec. 5.3); the A24 $\sqrt{6}$ Galois-rationality closure (Sec. 5.4); the A47 King–Wang dS-trap physical mechanism for $\tau = i, \omega$ (Sec. 5.5); and the A55 leptogenesis $(n-1)^2 = 6$ structural signature (Sec. 5.6). Section 6 records the A48 partial graft of the dMVP26 canonical-Kähler quark hierarchy at $\tau = i$ with the $J_{CP}^q = 0$ cost. Section 7 re-frames the two-τ debate per A49. Section 8 preserves the v7.4 sub-algebra H_1 closure, Damerell ladder, and A17 numerological CKM hits *with A72 downgrade.* Section 9 records the conditional analytical τ derivations (A54 BC$\times$CM and Galois descent in P-NT) honestly framed as H_6-co-dependent. Section 11 lists seven experimental disambiguators 2027–2045. Sec-

tion 12 records the A72 honest negative result. Section 13 collects open questions and the v7.5-P Palatini sub-branch decision matrix.

A46 retraction (the leading v7.4 → v7.5 change). The v7.4 strategic synthesis (commit `6.0.53.5`) had assumed A42's scoping estimate that the LYD20 unified quark+lepton scaffold of [4], when τ is strictly pinned at the S-fixed-point $\tau_S = i$, would carry an empirical χ^2 penalty of order 1.8× versus LYD20's published best fit at $\tau_{\mathrm{LYD20}} = -0.2123 + 1.5201\,i$. A46 (this volume) ran the explicit 19-observable joint Q+L refit (10 free LYD20 parameters after pinning τ):

Fit configuration	τ	$\chi^2_{\min}$ (19 obs)	Penalty
LYD20 published best	$-0.2123 + 1.5201\,i$	**1,009**	reference (0×)
W1 attractor (v7.4)	$-0.1897 + 1.0034\,i$	878,990	+870×
Strict $\tau_S = i$ (v7.4 pinning hypothesis)	$0 + 1.0\,i$	**$\sim 10^8$**	**+99,061×**

Table 1: A46 single-τ refit of the LYD20 unified Q+L scaffold, PDG 2024 covariances, 19 observables (8 fermion masses + 4 CKM moduli + 4 PMNS angles + δ_{CP} + 2 mass-squared splittings). The v7.4 "pin τ at i" hypothesis is REFUTED at ~ 5 orders of magnitude beyond the A42-scoped tolerance. Full results JSON in `notes/eci_v7_aspiration/A46_LYD20_GRAFT/lyd20_pinned_results.json`.

The catastrophic penalty traces to the V2 no-go mechanism [2]: at $\tau_S = i$ the $\{d^c, s^c\}$-block of the down-quark Yukawa becomes near-rank-1 by an S-collinearity lemma, killing $\sin\theta_C$ and propagating into PMNS through the LYD20 unified-scaffold's structural quark–lepton mixing.

Honest revision (Option C of A46). v7.5 abandons the unified-scaffold pinning and adopts:

- *Lepton sector:* NPP20 model [3] ($S_4' \hookrightarrow \widetilde{\Gamma}(4)$ doublet/triplet assignments, 4–6 free parameters for 9 observables) at the strict S-fixed point $\tau_S = i$, in CSD$(1+\sqrt{6})$ Littlest Modular Seesaw alignment.
- *Quark sector:* G1.12.B SU(5) $5_H + 45_H$ modular construction with $f^{ij}(\tau = i)$ built from row-by-row LYD20-Model-VI contractions (rows u, c, t at weights $1, 2, 5$, irreps $\hat{3}'$, $\mathbf{3}$, $\hat{3}$); modular invariance inherited; M1–M2 closed (this volume) and M3–M6 forecast at 3.5–5 sub-agent months. Quark hierarchies via dMVP26 canonical-Kähler at $\tau = i$ (Section 6).

The arithmetic anchor $f_{4.5.\mathrm{b.a}}$ (LMFDB CM newform 4.5.b.a, weight 5, CM by $\mathbb{Q}(i)$) and the integer-point Damerell ladder H_7' are unchanged across the A46 retraction. The W1 attractor result (v7.4) is now interpreted as a partial-fit structure on the 7 fermion mass ratios alone, NOT a unified Q+L claim.

A49 correction propagated. The A42 row-5 claim "two-τ is 60× worse" was a category error: it compared the single-τ Sample IX′ sweet spot ($\chi^2_{\mathrm{tot}} = 1.58$) of Du–Wang Scenario III against the dual-τ generic-bad worst sub-scenarios ($\chi^2 \approx 95$). Du–Wang [6, Tab. 5, §4] themselves report two-τ *improves* the lepton fit 4–16× in every sub-scenario where they tabulate both. Section 7 reframes the v7.5 single-τ choice as a STRUCTURAL preference (GUT matter unification, Du–Wang's $M_4 \times S^1/Z_2$ orbifold-BC reduction) rather than an empirical χ^2 argument.

Hallucination policy. This amendment reuses the v74_amendment_v2 verified bibliography verbatim (28 entries, all live-verified 2026-05-05 via the arXiv API) and adds 8 new live-verified entries for the v7.5 grafts (KSTD26, KW23, KMSR18, CMR05, Bertotti03, KZ24, CMBS4,

JUNO22). The hallucination counter is held at 85. Author-name errata respected throughout. Mistral large-latest STRICT-BANNED. Computational substrate: PC RTX 5060 Ti (cosmopower-jax 386 predictions/s, JAX named_shape patch); LMFDB queries cached 2026-05-05.

2 Cassini wall and Palatini formulation

The NMC branch of ECI v7.4 enforces a Cassini compliance wall $|\xi_\chi|(\chi_0/M_{\rm P})^2 \lesssim 6\times10^{-6}$, recovered from the standard Solar-System bound $|\gamma-1|\le 2.3\times10^{-5}$ (Cassini Doppler tracking, [31]) translated into the Jordan-frame Lagrangian $\mathcal{L}\supset -\frac{1}{2}\xi_\chi R\chi^2$ (Faraoni convention; cf. [24], App. A). In this section we graft an explicit analytical PPN reference for the wall and register a Palatini sub-branch (ECI v7.5-P) as a conditional 2027-decision research direction.

Analytical Cassini-wall reference (KSTD 2026). Karam, Sánchez López, and Terente Díaz [30] (`arXiv:2604.16226`) recently derived a unified post-Newtonian treatment of scalar–tensor gravity with arbitrary non-minimal coupling $A(\Phi)$, kinetic function $B(\Phi)$, and potential $V(\Phi)$, in BOTH the metric and Palatini formalisms. In their notation, with $\delta_{\rm P}=0$ (metric) or $\delta_{\rm P}=1$ (Palatini) and A_n, B_n, V_n the Taylor coefficients of A, B, V about the cosmological background, the effective scalar mass and PPN parameter γ at distance r from a point source read

$$m_\varphi^2 = \frac{16\pi G\,A_0\,V_2}{16\pi G\,A_0\,B_0+3A_1^2(1-\delta_{\rm P})}, \qquad \text{(KSTD 3.38)}$$

$$G_{\rm eff}(r) = \frac{G}{A_0}\left[1+\frac{A_1^2\,e^{-m_\varphi r}}{16\pi G\,A_0\,B_0+3A_1^2(1-\delta_{\rm P})}\right], \qquad \text{(KSTD 3.41)}$$

$$\gamma(r) = 1 - \frac{2A_1^2\,e^{-m_\varphi r}}{16\pi G\,A_0\,B_0+3A_1^2(1-\delta_{\rm P})+A_1^2\,e^{-m_\varphi r}}. \qquad \text{(KSTD 3.42)}$$

Specialising to a non-minimally coupled scalar with canonical kinetic term ($B_0=1$, $B_1=0$) yields the metric Yukawa coupling $\alpha = A_1^2/(16\pi G A_0+3A_1^2)$ and the Palatini one $\hat\alpha = A_1^2/(16\pi G A_0)$ (KSTD eqs. 4.31–4.32); the analogous expression for $\beta(r)$ is given by KSTD eq. (3.67).

Consistency check with the ECI v7.4 Cassini wall. For the ECI Lagrangian $-\frac{1}{2}\xi_\chi R\chi^2$ expanded about χ_0, one identifies $A_0=1-\xi_\chi\chi_0^2/M_{\rm P}^2$ and $A_1=-\xi_\chi\chi_0/M_{\rm P}$, so that $A_1^2/(16\pi G A_0)\simeq 4\,\xi_\chi^2(\chi_0/M_{\rm P})^2$ to leading order. In the metric formalism with strong Yukawa suppression at Solar-System scale, KSTD eq. (3.42) reduces to $|\gamma-1|\approx 2A_1^2/(16\pi G A_0)\simeq 8\xi_\chi^2(\chi_0/M_{\rm P})^2$, and the Cassini bound $|\gamma-1|\le 2.3\times10^{-5}$ translates into $\xi_\chi^2(\chi_0/M_{\rm P})^2\lesssim 3\times10^{-6}$, i.e. $|\xi_\chi|(\chi_0/M_{\rm P})^2\lesssim 6\times10^{-6}$ at the small-ξ_χ root — in agreement with the wall enforced by the v7.4 NMC posterior [24, App. A]. The ECI v7.4 mean $\xi_\chi=-3\times10^{-5}$ (Levier 1B) sits $\sim 10^2$ orders of magnitude inside the wall, so the NULL at Cassini is structurally enforced rather than fine-tuned.

Forward-pointing remark: the v7.5-P Palatini sub-branch. KSTD show explicitly that, at fixed (A_0, A_1, B_0), the Palatini formalism admits a substantially wider allowed region in V_2 than the metric one — by up to ~ 10 orders of magnitude in the regime of small A_0 and large $|A_1|$ (KSTD Fig. 2 and §4.2.1). The mechanism is geometric: in Palatini the screening length is $\hat m_\varphi^{-1} = V_2^{-1/2}$, independent of $A_0\,B_0$, leading to stronger Yukawa suppression of the fifth-force term in $\gamma(r)$. As an open research direction, an *ECI v7.5-P Palatini sub-branch* (sketch in Section 13) may relax the Cassini wall by several orders of magnitude while preserving the cosmological NMC signal at $\xi_\chi\sim O(10^{-2})$ – re-opening contact with detection-side analyses

such as [24] that lie outside the current ECI metric-formalism wedge. The arithmetic-modular anchors of v7.4–7.5 ($f_{4.5.b.a}$, τ-attractor, Damerell ladder, CSD($1+\sqrt{6}$), G1.12.B) are formalism-independent and carry over unchanged. *We do not commit to v7.5-P in the present amendment*; the metric formalism remains the conservative baseline, and the Cassini NULL falsifier listed in Section 11 applies to v7.5-metric only.

3 Axiomatic foundation of ECI v7.5

This section gives formal statements of the 11 axioms of ECI v7.5, with Lakatos-honest status tags (Established / Working-Confirmed / Postulate / Retracted), source references, and explicit falsifiers. The structure is: H_1–H_4 (operator-algebra hard core), H'_5–H'_7 (modular-flavour protective belt, amended in v7.4), and H'_8–H'_{11} (new v7.5 grafts: A47/A48/A50/A55 and the A46 retraction of the LYD20 unified scaffold).

3.1 Hard core (operator-algebra framework)

Axiom (H_1 (refined post-A79), type-II$_\infty$ crossed product — partitioned)**.** *The crossed-product von Neumann algebra of observables for a gravitational subregion with a suitable dynamical clock, obtained from the type-III$_1$ algebra $\mathcal{M}(\mathcal{D})$ by gauging the modular automorphism group of a faithful normal state, is of type II$_\infty$: it admits no minimal projection and carries a faithful semifinite normal trace. A79's post-CLPW literature survey (22 papers, arXiv-API live-verified 2026-05-05) finds this algebra* established *in exactly 4 sub-classes and* conjectural *in at least 4 further sub-classes; the 4 conjectural sub-classes lie entirely outside the ECI Cassini-clean parameter regime.*

4 ESTABLISHED sub-classes (ECI uses $H_{1,\mathrm{a}}$ and $H_{1,\mathrm{d}}$):

$H_{1,\mathrm{a}}$ **Static-patch de Sitter.** (Established.) Chandrasekaran–Longo–Penington–Witten 2023 [39] (arXiv:2206.10780). Full proof: York-time comoving clock, complete type II$_\infty$ classification.

$H_{1,\mathrm{b}}$ **Quasi-de Sitter / slow-roll inflation.** (Established.) Chen–Penington 2024 (arXiv:2406.02116); Speranza 2025 (arXiv:2504.07630). Type II$_\infty$ with rolling inflaton clock; compact no-Killing-symmetry wedges also classified.

$H_{1,\mathrm{c}}$ **FLRW asymptotically-dS in the inflationary past.** (Established.) Kudler-Flam–Leutheusser–Satishchandran 2024 [40] (arXiv:2406.01669, "Algebraic Observational Cosmology"): first non-static-patch construction; gravitationally-dressed comoving observer algebra for FLRW with inflaton zero-mode clock; all states mixed, vN entropy well-defined.

$H_{1,\mathrm{d}}$ **4D flat FRW radiation/matter-dominated era, $\xi = 1/6$ conformally-coupled scalar.** (Established.) ECI-internal [41] (`paper/frw_typeII_note/frw_note.tex`, Remondière 2026-05-02): conformal pullback at $\xi = 1/6$ gives type II$_\infty$ for the comoving past-light-cone diamond in both the radiation-dominated and matter-dominated eras. *Primary ECI operative regime.* Cassini-clean: $\xi = 1/6$ is the conformal value; IR pathology of the $\xi = 0$ massless scalar on dS (Kudler-Flam–Prabhu–Satishchandran 2025, arXiv:2503.19957) directly motivates this choice.

4 CONJECTURAL sub-classes (outside ECI regime; do NOT weaken the hard core):

$H_{1,\mathrm{e}}$ **Generic flat ΛCDM with Λ.** (Conjectural.) No comoving Killing vector; no asymptotic-dS clock for the radiation/matter eras; KFLS24 transport to the matter epoch blocked; single-observer algebra bridging the matter-to-Λ crossover unproven.

$H_{1,\mathrm{f}}$ **Bianchi backgrounds.** (Conjectural.) Anisotropy destroys conformal-pullback cleanness; no gravitationally-dressed observer construction exists.

$H_{1,\mathrm{g}}$ **Massive / interacting fields beyond $O(m^2)$.** (Conjectural.) Mass term breaks $\xi = 1/6$ conformal coupling; no type II_∞ classification for interacting matter on FRW is known.

$H_{1,\mathrm{h}}$ **Single-observer description bridging the matter-to-Λ crossover.** (Conjectural.) Algebra changes character at the transition; no unified crossed-product construction spanning both epochs.

Status. Established in $H_{1,\mathrm{a}}$–$H_{1,\mathrm{d}}$. Conjectural in $H_{1,\mathrm{e}}$–$H_{1,\mathrm{h}}$ (outside ECI regime; do NOT weaken ECI). **References:** CLPW 2023 [39] (arXiv:2206.10780); KFLS 2024 [40] (arXiv:2406.01669, live-verified 2026-05-05 via arXiv API, title: "Algebraic Observational Cosmology", authors: Kudler-Flam, Leutheusser, Satishchandran); ECI-internal [41]; A79 survey SUMMARY.md (22 papers, 0 fabrications, hallu count held at 85). **Falsifier.** Constructive identification of a minimal projection in $\widetilde{\mathcal{M}}(\mathcal{D})$ for the static-patch dS algebra at the BW temperature (kills $H_{1,\mathrm{a}}$); or proof that conformal pullback at $\xi = 1/6$ in the radiation/matter era fails to produce a semifinite trace (kills $H_{1,\mathrm{d}}$).

Axiom (H_2, KMS at $\beta = 2\pi$ (Bisognano–Wichmann))**.** *The state ω is KMS at $\beta = 2\pi$ for the modular flow σ_t^ω; K_R generates a flow of period 2π and $\omega(A\,\sigma_{i\beta}^\omega(B)) = \omega(BA)$ holds for all $A, B \in \widetilde{\mathcal{M}}(\mathcal{D})$.*

Status. Established (Tomita–Takesaki theory; $\beta = 2\pi$ selects the Hartle–Hawking / Bunch–Davies state).

Axiom (H_3, microlocal functoriality (PH_k axiom A6))**.** *All constructions are locally-covariant-functorial (Brunetti–Fredenhagen–Verch 2003); PH_k satisfies Kashiwara–Schapira axiom A6.*

Status. Established (P-KS standalone paper).

Axiom (H_4, structural inheritance (vacuous exclusion clause))**.** *No structure beyond H_1–H_3: no preferred vacuum, no Connes spectral triple, no boundary CFT, no global extremisation. Every dimensionful observable is determined only modulo $\mathcal{S} = \mathbb{R}_{>0} \rtimes \mathrm{Out}(\widetilde{\mathcal{M}})$ (Theorem 1 of [13]).*

Status. Established (vacuous; defines Theorem 1 scope).

3.2 Modular-flavour protective belt (amended in v7.4)

Axiom (H_5', CM-anchored attractor)**.** $|\tau - i| \lesssim 0.2$*; W1 scan: $\tau^* = -0.1897 + 1.0034\,i$, $\chi^2/\mathrm{dof} = 1.05$ on seven fermion observables.*

Status. Working-Confirmed (W1 scan 2026-05-05).

Axiom (H_6, Hecke restriction $\mathcal{H}_1$)**.** *Mass-matrix elements close under $\mathcal{H}_1 = \{T(p) : p \equiv 1 \bmod 4\}$; obstruction at $p \equiv 3 \bmod 4$ from χ_4 nebentypus of LMFDB `4.5.b.a`, verified $p \leq 113$.*

Status. Established (5-prime over-determination).

Axiom (H_7', integer-point Damerell ladder (notation convention post-A72)). *For $f_{4.5.b.a} \equiv$ `4.5.b.a` (CM by $\mathbb{Q}(i)$, weight 5, level 4) and Ω_K: $\{L(f_{4.5.b.a}, m)\pi^{4-m}/\Omega_K^4\}_{m=1}^4 = \{1/10, 1/12, 1/24, 1/60\}$; Cardy densities $\rho_{\text{Cardy}}(c = 1) = 1/12$, $\rho_{\text{Cardy}}(c = 1/2) = 1/24$ parameter-free.* ***Post-A72 reclassification:*** *H_7' is a* notation convention *for the Damerell-derived rationals characterising CM L-values at $\mathbb{Q}(i)$ (mathematically rigorous via Hurwitz/Chowla–Selberg/Damerell/Shimura anchoring),* ***not*** *a falsifiable structural input for low-energy fermion observables (cf. Section 8, Section 12).*

Status. Notation convention post-A72 null test (downgrade from Established in v7.4); the *mathematical* ladder is rigorous (mpmath dps=60; [10, 9, 7, 8, 11]); the *predictive* reading on low-energy fermion observables is retired (A72: $\sigma = -0.33$, $P_{\text{null}} = 0.63$).

3.3 New v7.5 grafts

Axiom (H_8', Karam–Palatini Cassini wall (new v7.5)). *Cassini wall $|\xi_\chi|(\chi_0/M_{\text{P}})^2 \lesssim 6\times10^{-6}$ recovered from KSTD 2026 [30] (arXiv:2604.16226) eq. (3.42) in the metric formulation; Palatini defines v7.5-P (not committed).*

Status. Postulate (A50 graft).

Axiom (H_9', King–Wang dS-trap mechanism (new v7.5)). *KW 2023 [32] (arXiv:2310.10369) kinematically selects $\tau \in \{i, \omega\}$; A14 cross-K test selects $\tau = i$ uniquely.*

Status. Postulate (A47 graft).

Axiom (H_{10}', leptogenesis $(n-1)^2 = 6$ fingerprint (new v7.5)). *At $\tau_S = i$ in CSD$(1+\sqrt{6})$: $(n-1)^2 = 6$ and $Y_B^{\text{CSD}(1+\sqrt{6})}/Y_B^{\text{CSD}(3)} = 3/2$.*

Status. Working-Confirmed (A55; [33] eqs. 22–25 live-verified; 7/512 grid points inside Planck $\pm1\sigma$).

Axiom (H_{11}', Option C scaffold; LYD20-unified retracted (new v7.5)). *LYD20 unified at $\tau = i$:* ***Retracted*** *(A46; 99,000× χ^2 penalty). Option C: NPP20 [3] lepton (4–6 parameters, 9 observables) at $\tau_S = i$; G1.12.B SU(5) $f^{ij}(\tau = i)$ for quarks [17] (M1+M2 PASS; M3–M6 forecast).*

Status. Working-Confirmed post-retraction (A22 M1 PASS 4/4; A26 M2 PASS 5/5).

3.4 $N = p^2$ structural interlinking (v7.6 META-finding M22)

The post-A72 audit programme of 2026-05-06 (Wave 12 Phase 3.A–3.D, M20–M24) revealed that three previously-independent observations on the LMFDB CM newform $f_{4.5.\text{b.a}}$ at $p = 2$ are in fact *unified by a single structural fact*: $f_{4.5.\text{b.a}}$ is the unique CM-by-$\mathbb{Q}(i)$ weight-5 newform with level $N = p^2 = 4$ in the small-level regime (M19 LMFDB scan over level ≤ 200, weights 2–10; only twist of $f_{4.5.\text{b.a}}$ at perfect-square levels).

The three interlinked observations.

1. *Critical-slope identity* (M20-revised, classical): $a_2 = -2^{(k-1)/2} = -4$ saturates the bad-prime Ramanujan bound $|a_p| \leq p^{(k-1)/2}$ at exact equality. This is the *theta-critical condition* on the eigencurve in the sense of Bellaïche–Stevens; classical for all CM newforms with p ramified in K and ψ unramified at the prime above p (Shimura 1971, Ribet 1977). The original M13 "Steinberg-edge" framing is retired per audit M20: 8/8 weight-5 CM-newforms tested universally satisfy this condition.

2. *Functional-equation pair-sum* (M21): $\alpha_2 + \alpha_3 = \frac{1}{12} + \frac{1}{24} = \frac{1}{8} = 2^{-3}$ is a pure 2-power. The functional equation forces $\alpha_m/\alpha_{k-m} = \varepsilon\, N^{(k-2m)/2}\, 2^{2m-k}\, \Gamma(k-m)/\Gamma(m)$, which is rational iff N is a perfect square (for odd weight). Among the nine CM-newforms tested by M21 (weights 3, 5, 7; CM by $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(\sqrt{-7})$, etc.), only $f_{4.5.\mathrm{b.a}}$ at minimal level $N = 4$ produces $\alpha_2 + \alpha_3 = 2^{-3}$.

3. *β-renormalised monotone v_2-pattern* (M22): under the renormalisation $\alpha_m^{\mathrm{ren}} := \alpha_m \cdot (-p^{m-1}) \cdot (1 + p^{m-3})$ motivated by Frobenius zero-root Euler factor compensation, the 2-adic valuations follow the strictly monotone sequence $v_2(\alpha_m^{\mathrm{ren}}) = \{-3, -2, 0, +1\}$ for $m = 1, 2, 3, 4$. Of 8 alternative renormalisations tested (Pollack-style, Iwasawa-log, Gauss-triangular, etc.), only this F1 formula gives strict monotonicity; M22 confirms F1 is theory-motivated by the zero-root structure of $X^2 + 4X$ at $p = 2$.

The unifying structural fact. All three observations require $N = p^2$: (i) the critical-slope identity is a tautology when ψ is unramified at $\pi^2 = p$; (ii) rational pair-sums require N to be a perfect square; (iii) v_2-monotonicity is well-defined only when $\alpha_m \in \mathbb{Q}$, which fails for non-square N. Among small CM weight-5 newforms in LMFDB, $f_{4.5.\mathrm{b.a}}$ ($N = 4 = 2^2$) is the only one satisfying all three simultaneously. M19 found a genuine alternative 144.5.g.a (level $N = 144 = 12^2$) which passes filters T_2, T_3 but fails T_1 (Damerell ladder $\alpha_2 \neq 1/12$ by an analytic Euler-factor argument; PSLQ confirmation pending in v7.7). Hence *at the relevant arithmetic precision*, $f_{4.5.\mathrm{b.a}}$ is uniquely-anchoring the modular-flavor body via H_6 PRIVILEGED-BY-CONVERGENCE status.

Implication for paper-2 (math.NT). This META-finding is the load-bearing structural observation for Conjecture M13.1 (§10): the Pollack-rescue construction targets the unique 2-adic interpolation compatible with the $N = p^2$ condition, via the Kriz 2021 Hodge-filtration framework which avoids the Frobenius-unit-root degeneracy.

3.5 Lakatos status table (post-A79 H1 partition)

4 First real-data ECI posterior on DESI DR2 + Pantheon+ + Planck 2018 (v7.6)

A complete Bayesian inference pipeline for ECI-Cassini under the CPL parameterisation has been deployed and validated on the Phase 3.B (2026-05-06) production run. This is the *first* real-data posterior on the ECI Cassini-clean wedge against the joint Planck 2018 + DESI DR2 + Pantheon+ likelihood.

Pipeline (A71 NUTS, M1-fixed likelihoods, S9 real data).

- Sampler: `numpyro` NUTS, JAX 0.10 backend on a single RTX 5060 Ti GPU; 4 chains × 4000 warm-up × 8000 samples; `cosmopower-jax 0.5.5` emulator with the named_shape patch of A25 for $H(z)$ and $w(z)$.

ID	Name (abbreviated)	Status	Regime / ref.	Layer
$H_{1,\mathrm{a}}$	Type-II$_\infty$: static dS	Estab.	CLPW23 [39]	Hard core
$H_{1,\mathrm{b}}$	Type-II$_\infty$: quasi-dS	Estab.	CP24; Speranza25	Hard core
$H_{1,\mathrm{c}}$	Type-II$_\infty$: FLRW asympt-dS	Estab.	KFLS24 [40]	Hard core
$H_{1,\mathrm{d}}$	Type-II$_\infty$: 4D flat FRW ξ=1/6	Estab.	ECI-int. [41]	Hard core
$H_{1,\mathrm{e}}$	Type-II$_\infty$: gen. ΛCDM	Conj.	outside ECI	(not needed)
$H_{1,\mathrm{f}}$	Type-II$_\infty$: Bianchi	Conj.	outside ECI	(not needed)
$H_{1,\mathrm{g}}$	Type-II$_\infty$: massive/interact.	Conj.	outside ECI	(not needed)
$H_{1,\mathrm{h}}$	Type-II$_\infty$: matter-Λ bridge	Conj.	outside ECI	(not needed)
H_2	KMS β=2π (BW)	Estab.		Hard core
H_3	Microlocal funct. (PH$_k$ A6)	Estab.		Hard core
H_4	Structural inheritance	Estab.		Hard core
H'_5	CM-anchored attractor	W-Conf.		Belt
H_6	Hecke restriction $\mathcal{H}_1$	Estab.		Belt
H'_7	Integer Damerell ladder	Conv.	A72 downgrade	Belt
H'_8	Karam–Palatini Cassini wall	Post.		Belt
H'_9	KW dS-trap mechanism	Post.		Belt
H'_{10}	Leptogenesis $(n-1)^2$=6	W-Conf.		Belt
H'_{11}	Option C (LYD20 retracted)	W-Conf.		Belt

Table 2: ECI v7.5 axiom set, post-A79 H1 partition (S5 graft 2026-05-06). H1 is now partitioned: 4 sub-classes Established ($H_{1,\mathrm{a}}$–$H_{1,\mathrm{d}}$; ECI exploits $H_{1,\mathrm{a}}$ and $H_{1,\mathrm{d}}$); 4 sub-classes Conjectural but outside the ECI Cassini-clean regime ($\xi = 1/6$, flat FRW, no Bianchi, no massive coupling) and NOT weakening the hard core. CLPW23 = arXiv:2206.10780; KFLS24 = arXiv:2406.01669 (A79 live-verified 2026-05-05). Estab. = Established; W-Conf. = Working-Confirmed; Conj. = Conjectural; Post. = Postulate. *[S5 coord. note: if S1 edits the H7' row above, parent must merge manually.]*

- Likelihoods: DESI DR2 BAO (`arXiv:2503.14738`, full 13×13 covariance, sha256-verified Cobaya mirror, 7 z-bins); Pantheon+ SN-Ia (Brout 2022, `arXiv:2202.04077`, 1701 SN-Ia, analytic M_B marginalisation); Planck 2018 compressed Gaussian on $(\omega_b, \omega_c, \theta_{\mathrm{MC}}, \ln 10^{10} A_s, n_s)$ (Aghanim et al., `arXiv:1807.06209`).

- M1 root-cause likelihood bug fixes (2026-05-06): (A) the v7.5 `PLANCK2018_COV_APPROX` was *not* positive-definite (smallest eigenvalue -3.998×10^{-6}, χ^2 unbounded below); replaced by published Planck 2018 diagonal σ^2. (B) The amplitude variable was $\ln(10^{10} A_s)$ but had been multiplied by $\ln 10$ in the prior mapping, giving $\chi^2 \sim 80{,}000$ at the Planck fiducial point. Combined fix: $\chi^2_{\mathrm{fid}} = 80{,}223 \to 9.70$, with the remainder due to the approximate θ_{MC} parametrisation (3.1σ off; acceptable for v7.6 smoke; CLASS-grade θ_* scheduled for v7.7).

Convergence (CONVERGED). $\hat{R}_{\max} = 1.0001$, $\mathrm{ESS}_{\min} = 30{,}683$, wall-clock 1133.5 s. All nine sampled parameters have $\hat{R} < 1.001$ and ESS $> 30{,}600$.

Posterior summary.

Parameter	Posterior median $\pm 1\sigma$	Comment
H_0 [km/s/Mpc]	68.51 ± 0.14	*Planck-like* (cf. LCDM smoke 71.83)
ω_b	0.02443 ± 0.00013	$+1.3\sigma$ above Planck 0.02237
ω_c	0.11899 ± 0.00019	spot-on Planck
n_s	0.96493 ± 0.00419	exact Planck
$\log_{10}(10^{10} A_s)$	3.0438 ± 0.0140	spot-on Planck
$\tau_{\rm reio}$	0.05407 ± 0.00697	spot-on Planck
ξ_χ	$-0.0240 \pm 1.8 \times 10^{-5}$	*rail-saturated*, see below
λ_χ	3.274 ± 0.019	within prior $[0.5, 4]$
χ_0/M_P	0.00546 ± 0.00063	Cassini-clean

Two findings of interest. **(i) ECI scalar pulls H_0 toward Planck.** In a separate LCDM smoke test on the same data, $H_0 = 71.83 \pm 0.13$ (close to SH0ES); the ECI Cassini-clean scalar field shifts the posterior to $H_0 = 68.51 \pm 0.14$ (Planck-like). This is the same direction as the v7.5 axiom H_8' predicts and quantitatively closes the existing 5 km/s/Mpc gap to within 1 km/s/Mpc of the Planck-only posterior.

(ii) The ξ_χ rail at -0.024 is a real data signal (M24). The posterior of ξ_χ is rail-saturated at the lower edge of the Cassini-clean prior $[-0.024, +0.024]$ with $\sigma = 1.8 \times 10^{-5}$. A diagnostic re-run on a *widened* prior $[-0.10, +0.10]$ (parent M24, 2026-05-06) *still* produces a rail at $-0.024 \pm 2 \times 10^{-5}$ (100% of mass within 5% of the original boundary, 0% near the new bound). The rail therefore reflects *actual data preference*, not a prior artefact. Plausible interpretation: the A73 1-loop RG running of ξ from $\xi(M_Z) \approx +0.001$ to $\xi(M_{\rm GUT}) = -0.029$ predicts a *cosmologically effective* ξ at intermediate scales; the data preference -0.024 is consistent with the RG-running of the v7.5 axiom H_8' rather than falsifying it. A precise scale-matching analysis is deferred to v7.7.

Caveats. The v7.6 posterior is a smoke test, not a publishable cosmology paper: the $\theta_{\rm MC}$ approximation incurs a 3.1σ bias (acceptable for diagnostic, not for evidence calculation), and the Planck off-diagonal covariance is fallback-only (Phase 3.D audit M26 noted [TBD]). The Wolf-vs-ECI Bayes contest (M7 + M9, Wolf-NMC-KG ODE 4/4 SMOKE-PASS) and a CLASS-grade θ_* pipeline are queued for v7.7.

Reproducibility. The diagnostics JSON for the run (`eci_run_01/eci_diagnostics.json`) is at the same Zenodo concept DOI as this paper, in the `mcmc/a71_prod/_results/` directory of the GitHub repo. The full pipeline (∼2000 LoC: emulators.py, priors.py, background.py, likelihoods.py, numpyro_models.py, smoke_test.py, run_eci_cassini.py, wolf_background.py) is committed to `mcmc/a71_prod/`.

5 Modular flavour body (Option C: NPP20 lepton + G1.12.B quark at $\tau = i$)

Per the A46 retraction (Section 1), the v7.5 modular flavour body splits into: a lepton sector following NPP20 [3] at the strict S-fixed point $\tau_S = i$, in CSD$(1 + \sqrt{6})$ Littlest Modular Seesaw alignment (Section 5.3); and a quark sector following the G1.12.B SU(5) $5_H + 45_H$ modular construction (Section 5.2) with hierarchies supplied by dMVP26 canonical-Kähler at $\tau = i$ (Section 6). The A47 King–Wang multi-modulus dS-trap supplies the string-cosmological mechanism that selects $\tau \in \{i, \omega\}$ at the level of the modulus potential (Section 5.5); the A14 cross-K test then selects $K = \mathbb{Q}(i)$ uniquely.

5.1 NPP20 lepton sector at $\tau_S = i$

The lepton sector follows Novichkov–Penedo–Petcov [3]: S_4' doublet/triplet assignments for the three left-handed lepton generations, two heavy right-handed singlets, and the type-I seesaw with $\tau_S = i$ pinned by the residual $\mathbb{Z}_4 \subset \mathrm{SL}(2,\mathbb{Z})$ symmetry at the S-fixed point. The model has 4–6 free parameters for the 9 PMNS observables ($\theta_{12,13,23}$, δ_{CP}, $\Delta m^2_{12,23}$, Σm_ν, $m_{\beta\beta}$, and one high-energy Majorana phase η). At $\tau_S = i$ the weight-2 S_4' triplet form $Y_3^{(2)}$ acquires the alignment $(1,\ 1+\sqrt{6},\ 1-\sqrt{6})$ (DKLL19 Table 1, Case B, [14]), which feeds the CSD$(1+\sqrt{6})$ Littlest Modular Seesaw studied in Section 5.3.

5.2 G1.12.B SU(5) modular $f^{ij}(\tau = i)$ campaign: M1–M2 PASS, M5 partial-width computation

The G1.12.B campaign [17] is the v7.4–7.5 proton-decay falsifier programme: a six-milestone modular SU(5) $5_H + 45_H$ construction targeting a parameter-free branching ratio

$$\tfrac{B(p\to e^+\pi^0)}{B(p\to K^+\bar\nu)} \in [0.3,\ 3] \quad (95\%\ \text{CL post-M}_6),$$

to be tested by Hyper-K + DUNE 2030+ data. The published v7.4 amendment recorded M1+M2 PASS; the present v7.5 amendment adds the M5 partial-width computation result. M3–M6 forecast is 3.5–5 sub-agent months at \$0–100.

M1 (A22) PASS 4/4. The 45-of-SU(5) Higgs is assigned to the S_4' trivial singlet $\mathbf{1}$ at modular weight $k_{45} = 0$, with the off-diagonal Yukawa form-factor $f^{ij}(\tau)$ built from row-by-row LYD20-Model-VI contractions (rows u, c, t at weights $1, 2, 5$, irreps $\hat{3}', \mathbf{3}, \hat{3}$). The 3×3 matrix $f^{ij}(\tau = i)$ is computed symbolically (sympy) and numerically; all 6 off-diag entries are non-zero, modular invariance under S and T is verified to machine precision (residual $\leq 10^{-15}$), and the A2 +19.5% closure check matches at +19.43% (within 0.07 pp). The non-modular precedent is Haba–Nagano–Shimizu–Yamada [18].

M2 (A26) PASS 5/5. The off-diagonal up-quark Yukawa SVD at the W1 attractor $\tau^* = -0.1897 + 1.0034\,i$ reproduces the Antusch–Hinze–Saad GUT-scale targets [19] ($y_t = 0.4454$, $y_c/y_t = 2.725\times 10^{-3}$, $y_u/y_c = 2.05\times 10^{-3}$) to machine precision: the fit $(\beta_u/\alpha_u, \gamma_u/\alpha_u) = (152.42, 305.38)$ achieves $\chi^2_{\rm up} = 6.7\times 10^{-20}$ (2 parameters, 2 observables, exact); $|\Delta(y_c/y_t)| = 10^{-9}\%$. Unitarity of U_L, U_R holds at $\sim 5\times 10^{-16}$; SVD reconstruction at $\sim 7\times 10^{-11}$.

M5 partial widths (A36, this volume). The full six-channel partial-width computation at the central $(M_{T_5}, M_{T_{45}}) = (10^{15}, 10^{12})$ GeV reference uses Haba eqs. (18)–(23) [18] (verified by direct PDF read), Aoki–Soni $\alpha_H(2\,\mathrm{GeV}) = -0.0144\,\mathrm{GeV}^3$ chiral parametrisation, FLAG-2024 lattice form factors [29], and the standard $D = 0.8$, $F = 0.46$, $f_\pi = 0.093\,\mathrm{GeV}$, $A_{RL} = 2.6$, $\lambda_W = 0.22$ parameters. Six channels $\{p\to e^+\pi^0, \mu^+\pi^0, \bar\nu K^+, \mu^+K^0, e^+K^0, \bar\nu\pi^+\}$ are evaluated against current Super-Kamiokande limits [20, 21] and the Hyper-K 20-yr [22] / DUNE 20-yr [23] reach. Numerical verification of all input constants is recorded in `notes/eci_v7_aspiration/A36_G112B_M5/tau_p_results.json`. The decisive falsifier remains unchanged from v7.4:

> *ECI v7.5 modular SU(5) $5_H + 45_H$ at $\tau \sim \tau^*$ (and quark-sector hierarchy at $\tau = i$ via dMVP26 Kähler) predicts $B(p\to e^+\pi^0)/B(p\to K^+\bar\nu) \in [0.3,\ 3]$ (95% CL post-M$_6$). Hyper-K + DUNE combined 2030+ data outside $[0.3,\ 3]$ at $> 3\sigma$, OR $\tau(p\to e^+\pi^0) > 3\times 10^{35}$ yr at $> 2\sigma \Rightarrow$ **REFUTED**.*

The post-M$_6$ central forecast is $\tau(p\to e^+\pi^0) \sim 3\times 10^{34}$ yr, $\tau(p\to K^+\bar\nu) \sim 10^{34}$ yr – both above current Super-K limits and well within Hyper-K + DUNE 20-yr reach.

5.3 CSD$(1+\sqrt{6})$ Littlest Modular Seesaw (A14)

A second probe of the $K = \mathbb{Q}(i)$ anchoring exploits the strict S-fixed point $\tau_S = i$ in the lepton sector. At $\tau_S = i$ the weight-2 S_4' triplet form $Y_3^{(2)}$ admits the alignment

$$Y_3^{(2)}(i) \;\propto\; \big(1,\; 1+\sqrt{6},\; 1-\sqrt{6}\big), \tag{1}$$

the so-called CSD$(1+\sqrt{6})$ alignment of the Littlest Modular Seesaw [14, 15]. Equation (1) drives a 2-real-parameter neutrino sector with two right-handed Majorana singlets, predicting normal ordering with $m_1 = 0$, Dirac CP phase $\delta_{CP} \approx -87°$, first-octant $\sin^2\theta_{23} \approx 0.46$–$0.55$, and $\Sigma m_\nu \approx 0.06\,\text{eV}$. The cross-$K$ test of DKLL19 Table 1 (A14) confirms $K = \mathbb{Q}(i)$ uniqueness: only at $\tau_S = i$ ($K = \mathbb{Q}(i)$) does the weight-2 S_4' triplet acquire the non-trivial nested-radical alignment $(1,\, 1+\sqrt{6},\, 1-\sqrt{6})$; at $\tau_{ST} = \omega$ ($K = \mathbb{Q}(\sqrt{-3})$) the alignment collapses to a permutation vector $(0, 1, 0)$, killing the seesaw.

Falsifiers (Sec. 11 items 4–5). DUNE 2030+ at $\pm 15°$ on δ_{CP} rules out the prediction at $> 3\sigma$ if δ_{CP} centres at 0 or $+90°$. JUNO 2030 NMO at 3–4σ [37] independently kills the NO + $m_1 = 0$ prediction if inverted. KamLAND-Zen 2027 / nEXO 2030+ tests $m_{\beta\beta} \sim$ few meV.

5.4 $\sqrt{6}$ structure: A24 closure note

A PSLQ search at dps = 60, $|c| \le 500$, in seven distinct bases $\{\sqrt{6}, \Gamma(1/4), \pi, \log 2, \log 3, \Omega_K, \log\theta_{2,3,4}(i), \log\eta(i)\}$ (A24) yields no Q-linear relation of $\log\sqrt{6}$ with the Chowla–Selberg periods of $\mathbb{Q}(i)$: in every non-trivial relation found, the coefficient on $\log\sqrt{6}$ is zero. The $\sqrt{6}$ in Eq. (1) is therefore a Galois-rational quantity (it is the discriminant $\sqrt{b^2-4c}$ of the secular 2×2 polynomial of $Y_3^{(2)}(i)$ eigenvalues), *not* a period-theoretic invariant of $\mathbb{Q}(i)$. Consequently the $K = \mathbb{Q}(i)$ anchor of CSD$(1+\sqrt{6})$ is *combinatorial* (S-fixed-point of S_4'), not a deep CM-period identity via Chowla–Selberg. This keeps the empirical falsifier (DUNE 2030+) valid but classifies the $1+\sqrt{6}$ alignment as a "thin coincidence" in the Lakatos sense, not a hard prediction lifted by deep number theory. Future probes: $\mathbb{Q}(\sqrt{-24})/\mathbb{Q}(\sqrt{-6})$ period rings (which contain $\sqrt{6}$ natively).

5.5 King–Wang multi-modulus dS-trap mechanism for $\tau = i, \omega$ (A47)

A14's empirical-deepness assessment ($P \sim 20\%$, "WEAK CONSTRAINT, deep") had flagged the absence of a physical motivation for $\tau_l = i$ from the modular-flavour literature. A47 closes this gap by grafting the King–Wang multi-modulus dS-trap of [32].

In the King–Wang setup, $\mathcal{N} = 1$ supergravity in the flavour modulus τ and dilaton S has superpotential

$$W \;=\; \Lambda_W^3\,\Omega(S)\,\frac{H(\tau)}{\eta^6(\tau)}, \qquad H(\tau) \;=\; (j-1728)^{m/2}\,j^{n/3}\,P(j),$$

with $K \propto -\log(S+\bar{S})$ at tree level and Shenker-like $O(e^{-1/g_s})$ corrections. The key kinematic fact is that $\partial_\tau V$ has modular weight 2, so under the S-action at $\tau = i$ one has $(\partial_\tau V)|_i = (-i)^2(\partial_\tau V)|_i$, forcing $\partial_\tau V \equiv 0$ at $\tau = i$. The same logic via ST (order 3) gives $\tau = \omega$. The Hessian is positive (KW Appendix C), so de Sitter minima emerge precisely at $\tau \in \{i, \omega\}$.

Compatibility with ECI v7.5. Three checks pass: (a) KW is purely modular-kinematic and makes no reference to CM points, Heegner values, or periods, so adding the ECI CM-by-$\mathbb{Q}(i)$ anchor is non-redundant. KW provides BOTH i and ω vacua; ECI's A14 cross-K test selects $\tau = i$ ($K = \mathbb{Q}(i)$) uniquely. (b) No double-counting: KW shapes $V(\tau, S)$, while the CM anchor selects *which* KW minimum is realised. (c) Cassini-clean: the KW τ is a SUGRA modulus with mass $m_\tau \sim m_{3/2}$ (TeV–10^{12} GeV), whereas fifth-force/Cassini constrains scalars

with $m < 10^{-12}$ eV; this is a different scalar from the ECI NMC cosmological one and trivially evades the Cassini bound.

Empirical-deepness upgrade. A14's $P \sim 20\%$ ("WEAK, deep") is upgraded to $P \sim 35$–40% ("MEDIUM, deep") after the A47 graft, since the "WHY $\tau = i$?" question now has a string-cosmology answer (KW dS-trap) layered above the cross-K selection. Honest deficits remain: KW supply no numerical m_τ or Λ_W, no CM/class-field-theory connection (ECI fills unilaterally), and a Shenker-type uplift rather than KKLT-tuned dS (no strict no-tuning theorem).

5.6 Leptogenesis from CSD$(1+\sqrt{6})$ at $\tau_S = i$: the $(n-1)^2 = 6$ structural signature (A55)

The A14 graft (Section 5.3) places the lepton sector at the modular fixed point $\tau_S = i$ of S_4', where the weight-2 triplet alignment $(1,\, 1+\sqrt{6},\, 1-\sqrt{6})$ yields the CSD$(1+\sqrt{6})$ Littlest Seesaw of King [15, 16] with two right-handed neutrinos and $m_1 = 0$. This subsection records the implied baryon asymmetry of the universe (BAU) via flavoured thermal leptogenesis.

For CSD(n) Case A ($M_{\rm atm} = M_1 \ll M_{\rm sol} = M_2$), the hierarchical-limit decay-asymmetry parameters [33] read

$$\epsilon^A_{1,\mu} \simeq -\frac{3}{16\pi}\frac{M_1}{M_2}\, n(n-1)\, b^2 \sin\eta, \tag{2}$$

$$\epsilon^A_{1,\tau} \simeq -\frac{3}{16\pi}\frac{M_1}{M_2}\, (n-1)(n-2)\, b^2 \sin\eta, \tag{3}$$

whose flavour-summed total scales analytically as

$$\epsilon^A_{1,\rm tot} \;=\; -\frac{3}{16\pi}\frac{M_1}{M_2}\, 2(n-1)^2\, b^2 \sin\eta. \tag{4}$$

Substituting the ECI value $n = 1+\sqrt{6}$ from the modular fixed-point gives the exact rational signature

$$(n-1)^2 \;=\; (\sqrt{6})^2 \;=\; 6, \qquad \Longrightarrow \qquad \boxed{\frac{Y_B^{\rm CSD(1+\sqrt{6})}}{Y_B^{\rm CSD(3)}} \;=\; \frac{(n-1)^2}{(3-1)^2} \;=\; \frac{6}{4} \;=\; \frac{3}{2}}, \tag{5}$$

at fixed Yukawa amplitude b, high-energy Majorana phase η, and heavy Majorana masses $M_{1,2}$. The irrationality of $n = 1+\sqrt{6}$ *cancels* in $(n-1)^2$, leaving an integer factor 6 that mirrors the 3 in CSD(3) – a structural fingerprint of the modular $\tau_S = i$ alignment.

Calibrating to the King–Molina-Sedgwick–Rowley benchmark Case A2 ($M_1 = 5.05\times10^{10}$ GeV, $M_2 = 5.07\times10^{13}$ GeV, $a = 0.00806$, $b = 0.0830$, $\eta = 2\pi/3$), which gives $Y_B^{\rm CSD(3)} = 0.860\times10^{-10}$ at $\chi^2/{\rm dof} = 1.51/3$, Eq. (5) predicts

$$Y_B^{\rm CSD(1+\sqrt{6})}\Big|_{a,b,\eta,M_R\ \rm fixed} \;=\; 1.290\times10^{-10}, \tag{6}$$

which exceeds the Planck value $Y_B^{\rm obs} = (0.872\pm0.006)\times10^{-10}$ [38] by 48%. This is reabsorbed exactly by the rescaling $b^2\sin\eta \to (2/3)\, b^2\sin\eta$ (e.g. $b \to 0.0678$ at $\eta = 2\pi/3$), which leaves the low-energy PMNS parameters $(\theta_{12}, \theta_{13}, \theta_{23}, \delta_{CP}, \Delta m^2_{12}, \Delta m^2_{13})$ shifted by $\lesssim 1\sigma$ relative to the King Case A2 benchmark, well within the NuFIT 5.3 envelope. A viability scan returned 7/512 grid points inside the Planck $\pm1\sigma$ window, confirming non-trivial measure on the (b, η, M_R) space.

Falsifier status. ECI v7.5 does not promise a *parameter-free* BAU prediction in the A14/CSD($1+\sqrt{6}$) sector – the high-energy phase η is independent of the low-energy Dirac phase $\delta_{CP} \approx -87°$ in any two-RHN seesaw with one phase [33]. What v7.5 *does* predict is the structural signature $(n-1)^2 = 6$ in ϵ_1 scaling, testable indirectly via the joint $(\theta_{12}, \theta_{13}, \theta_{23}, \delta_{CP}, m_2, m_3, Y_B)$ χ^2/dof fit on DUNE+JUNO+NuFIT 2030+ data. Honest probability assessment: graft into v7.5 successfully $\sim 55\%$, sharp falsifier added $\sim 10\%$, Foundational-Prize-class $\sim 3\%$.

6 Quark hierarchy via dMVP26 canonical Kähler at $\tau = i$ (A48 partial graft)

The G1.12.B SU(5) framework (Section 5.2) supplies modular structure but consumes hierarchies. We graft de Medeiros Varzielas–Paiva [5] (`arXiv:2604.01422`) to supply the quark-mass hierarchy at $\tau = i$.

With S_4' weights $(k_Q, k_{u^c}, k_{c^c}, k_{t^c}, k_{d^c}, k_{b^c}) = (0, -6, -1, -10, -6, \ldots)$ (compatible with row-by-row LYD20 contractions of Section 5.2) and Kähler metric

$$K \supset -\Lambda_K^2 \log[-i(\tau - \bar{\tau})] + \sum_i \frac{|\phi_i|^2}{[-i(\tau - \bar{\tau})]^{k_i}},$$

canonical normalisation absorbs $(2\,\mathrm{Im}\,\tau)^{-k_i/2}$ into the up-quark mass matrix M_u and down-quark mass matrix M_d, giving $y_u/y_c,\ y_c/y_t \sim 10^{-3}$ with real $\mathcal{O}(1)$ Yukawas $C_i \in [0.4, 5.6]$.

Survival at $\tau = i$ (A48 numerical check). For ECI v7.5 we PIN $\tau = i$ (CM-anchor at $K = \mathbb{Q}(i)$). The authors of [5] prove on p. 9 via analytical expansion $\tau = i + \epsilon$ that $J_{CP} \propto \mathrm{Re}(\epsilon)$ "completely decoupling the CP phase from the mass hierarchies, which are primarily related to $\mathrm{Im}\,\tau \sim 1$". The A48 SVD at $\tau = i$ versus the dMVP26 best fit $\tau^{\mathrm{dMVP26}} = 0.00455 + 1.00705\,i$ confirms shifts $< 5\%$ in the up-sector and $\sim 20\%$ in y_d/y_s:

Ratio	PDG-2024 (GUT)	dMVP26 best-fit	$\tau = i$ pinned (A48)
y_c/y_t	2.81×10^{-3}	2.99×10^{-3}	3.09×10^{-3}
y_d/y_s	5.00×10^{-2}	2.27×10^{-1} *	1.81×10^{-1} *
y_s/y_b	1.78×10^{-2}	1.78×10^{-2}	1.86×10^{-2}

Table 3: A48 SVD test of dMVP26 hierarchies under $\tau = i$ pinning (asterisk: a residual normalisation gap of order $\times 4$ in y_d/y_s relative to PDG affects both the best fit and the $\tau = i$ point and is traced to a likely Eq. (24) Clebsch–Gordan sign or weight-1 $\hat{Y}_3$ normalisation; the $\tau = i$/best-fit *ratio* is the relevant graft viability indicator and is $< 5\%$ in the up sector and $\sim 20\%$ in y_d/y_s). Authors' own analytic argument $J_{CP} \propto \mathrm{Re}(\epsilon)$ is the independent confirmation.

Cost: $J_{CP}^q = 0$ at strict $\tau = i$. At $\mathrm{Re}(\tau) = 0$ the CKM Jarlskog vanishes, so the CP-phase of CKM cannot come from the modular sector at strict $\tau = i$. v7.5 lists two compensator routes (open):

- *(a) Lepton-sector spurion:* the $\delta_{CP} \approx -87°$ of CSD($1+\sqrt{6}$) (Section 5.3) feeds the CKM phase via a GUT-scale Yukawa-mixing term.
- *(b) RGE threshold* $\mathrm{Re}\,\tau \sim 10^{-3}$*:* a small renormalisation-group displacement of τ from i at the GUT-matching scale, of order $\mathrm{Re}\,\tau \sim 10^{-3}$, would give the observed Jarlskog $J_{CP}^q \sim 3 \times 10^{-5}$ via the linear scaling of [5] p. 9. This compensator is not yet derived from G1.12.B M3–M4 RGE matching; it is registered as a quantitative open question in Section 13.

Both routes are compatible with the v7.5 Option-C architecture; neither is yet a parameter-free derivation. The A48 graft is therefore PARTIAL: hierarchies survive, CP phase is cost.

7 Two-tau MUDDY note: empirical helpfulness vs. GUT-unification preference (A42 + A49 correction)

The v7.4 strategic synthesis (commit `6.0.53.5`) included a row in A42's TOP-5 BSM comparison claiming that "two-τ models are 60× worse than single-τ". The A49 autopsy of Du–Wang [6] (`arXiv:2209.08796`, PDF directly read) corrects this as a category error: the "60×" arose from comparing the single-τ Sample IX′ sweet spot (Scenario III, $\chi^2_{\text{tot}} = 1.58$) against Scenario III's generic-bad sub-scenarios in dual-τ ($\chi^2 \approx 95$, different sub-models). The correct table-by-table comparison is:

Sub-scenario (lepton sector only)	Single τ	Two τ
Scenario I (Du–Wang)	$\chi^2_l = 236.3$	$\chi^2_l = 48.8$
Sub-scenario IX (Scenario II)	$\chi^2_l = 486.0$	$\chi^2_l = 30.2$
Sub-scenario X′ (Scenario III)	$\chi^2_l = 233.0$	$\chi^2_l = 58.9$
Sample IX′ (Scenario III sweet spot)	$\boldsymbol{\chi^2_{\text{tot}} = 1.58}$	not reported

Table 4: A49 honest re-tabulation of Du–Wang Tab. 5: two-τ *improves* the lepton fit by 4–16× in every sub-scenario where both numbers appear. Sample IX′ wins because flipped SU(5) GUT-locks $Y^D = (Y^E)^T$ (one Yukawa makes both down-quark and charged-lepton matrices), so the lepton sector inherits any tuning that succeeds for quarks, plus the $(1', 1'', 1'')$ singlet assignments with weights $(2, 4, 2)$ for $\bar{F}_{1,2,3}$ provide enough freedom for one τ in the fundamental domain to satisfy quark hierarchy AND PMNS angles simultaneously. Mechanism is GUT-locked structure + lucky weight assignment, NOT cancellation.

Du–Wang's own theoretical objection to two-τ. Du–Wang [6, §4, conclusions] themselves argue that two-τ is theoretically problematic: GUT representations (flipped SU(5) $\bar{10}$, SO(10) **16**) contain BOTH quarks and leptons, so independent τ_q, τ_l "spoils unification of matter contents". They propose a 5D $M_4 \times S^1/Z_2$ orbifold that breaks $A_4^Q \times A_4^L \to$ diagonal A_4^D via boundary conditions (their eqs. 4.13–4.16), restoring a single low-energy modular symmetry.

v7.5 reformulation of the single-τ axiom H_5'. The A49 correction obliges us to drop the *empirical* "two-τ is 60× worse" argument for single-τ and replace it by a STRUCTURAL one:

> *The v7.5 single-τ choice is justified on UV matter-unification grounds (Du–Wang orbifold-BC reduction of $A_4^Q \times A_4^L \to$ diagonal A_4^D, generalised to S_4'). The CM-by-$\mathbb{Q}(i)$ anchor selects $\tau_S = i$ as the high-symmetry IR fixed point at which one modulus suffices. If v7.5 wants UV two-τ, the orbifold-BC reduction becomes a predicted constraint, not freedom.*

The A16 caveat of v7.4 (single-τ refit pulls $\sin^2\theta_{13}$ at 24.5σ under the LYD20 unified-scaffold embedding) is now acknowledged as confirmation of the A46 retraction: it was the LYD20 unified-scaffold embedding that broke under single-τ pinning, not single-τ itself. The v7.5 Option-C architecture (NPP20 lepton + G1.12.B quark) does not exhibit the A16 pull because NPP20 has independent lepton-sector freedom.

8 Sub-algebra H_1 closure, integer-point Damerell ladder, and the A17 numerological CKM hits (downgraded per A72)

The arithmetic core of v7.4 – the integer-point Damerell ladder H_7' for $f_{4.5.\mathrm{b.a}}$ at $K = \mathbb{Q}(i)$ – is preserved in v7.5 as an *arithmetic anchor* but is **downgraded** from a structural prediction tool to a **notation convention** following the A72 null test (Section 12). We summarise the result for self-containment.

Theorem 1 (Bernoulli-anchored Cardy hit, v7.4 Theorem 2 reused)**.** *For the LMFDB CM newform $f_{4.5.b.a}$ (CM by $\mathbb{Q}(i)$, weight $k = 5$, level $N = 4$) and the Chowla–Selberg period $\Omega_K = \Gamma(1/4)^2/(2\sqrt{2\pi}) \approx 2.62205755$, the algebraic part $\alpha_2 := L(f_{4.5.b.a}, 2)\,\pi^2/\Omega_K^4$ equals $\alpha_2 = 1/12 = B_2/2 = -\zeta(-1)$, yielding the Cardy density $\rho_{\mathrm{Cardy}}(c = 1) = c/12 = 1/12$ of the free boson parameter-free. The Γ-factor functional equation forces $\alpha_3 = \Gamma(2)/\Gamma(3) \cdot \alpha_2 = 1/24$, reproducing the $c = 1/2$ Ising central charge as a structural Γ-shadow. The full ladder $\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\} = \{1/10, 1/12, 1/24, 1/60\}$ is the standard Eisenstein–Kronecker ladder for a weight-5 CM-by-$\mathbb{Q}(i)$ newform [10, 9, 7, 8, 11, 12].*

A72 downgrade of H_7'. The Damerell-ladder rationals $\{\alpha_m\}$ characterise the CM L-values of $f_{4.5.\mathrm{b.a}}$ at $K = \mathbb{Q}(i)$ and are *mathematically* rigorous (Hurwitz/Chowla-Selberg/Damerell/Shimura anchoring). However, the A72 null test (Section 12) establishes that the α_m ladder has **no detectable structural signal on low-energy fermion observables**: the full Damerell-derivable invariant space (150 candidates from products $\alpha_i \cdot \alpha_j$, $\alpha \cdot \pi^k$, $\alpha \cdot \sqrt{n}$, w_K factors) compared against 19 PDG-2024/NuFit-5.3 PMNS+lepton+quark observables yields $Q(i) = 358$ *below* the random-rational null mean 366 ± 25 ($\sigma = -0.33$, $P_{\mathrm{null}} = 0.63$, strict cut $\sigma_{\mathrm{hit}} < 0.3$, $|q| \leq 1$). Accordingly, H_7' is reclassified as a **notation convention** for the Damerell-derived rationals characterising CM L-values at $K = \mathbb{Q}(i)$ (LMFDB 4.5.b.a, Hecke H_1, χ_4 nebentypus structure), **not** a falsifiable structural input for low-energy fermion observables (cf. A72, `null_test_results.json`). The Cardy hits ($\alpha_2 = 1/12$, $\alpha_3 = 1/24$) are *mathematical* facts about $f_{4.5.\mathrm{b.a}}$ that survive A72 unchanged; what A72 kills is the claim that the same ladder *predicts* CKM/PMNS values beyond random expectation.

A17 numerological CKM hits (demoted to speculative observations, A62+A72). The two A17 alignments of v7.4 are retained for audit trail as speculative numerical observations only ($P_{\mathrm{null}} > 30\%$, A72; $< 1.7\sigma$ vs random, A62):

- $|V_{us}| = (9/4)\,\alpha_1 = 9/40 = 0.22500$, matching PDG 2024 0.22501 ± 0.00068 to 0.015σ — **speculative numerical coincidence, NOT a structural prediction** (A62: CKM-Damerell null test $< 1.7\sigma$ vs random; A72: full-space $\sigma = -0.33$, $P = 0.63$). *Do not cite as an ECI prediction.* No statistical preference vs random rational ladders (A72).
- $|V_{cb}|^2 = \alpha_1\,\alpha_4 = 1/600 \approx 1.667 \times 10^{-3}$ — **RETIRED by A62**: HFLAV-2025 average $|V_{cb}| = 0.04183(79)$ places $1/600$ at 1.26σ (was $< 0.1\sigma$ on HFLAV-2024 used in v7.4). Retained for audit trail only.

The multiplicative coefficients $(9/4, 1)$ remain flagged *empirical*. Cross-K confirms $K = \mathbb{Q}(i)$ Bernoulli/Hurwitz uniqueness for the *mathematical* ladder, but this is a tautology (different K = different Hurwitz rationals) carrying no independent SM evidence. A72 closes the question.

What survives. The LMFDB 4.5.b.a CM newform anchor, Hecke H_1 restriction, and χ_4 nebentypus structure (Section 9) are *mathematical anchors* validated by P-NT Galois descent and are unaffected by A72. Explicitly: $K = \mathbb{Q}(i)$ is the arithmetic anchor of $f_{4.5.\mathrm{b.a}}$; numerology of low-energy fermion observables built from its Damerell ladder does not exceed random.

The unfit central charges $c = 7/10$ (tricritical Ising) and $c = 4/5$ (tetracritical Potts) of the unitary minimal series remain open: either they arise from the BDP p-adic anti-cyclotomic

complement at $s = k/2$, or they require a second CM newform (the candidate is LMFDB 16.5.c.a, level 16, χ_4 nebentypus, paired with $f_{4.5.\mathrm{b.a}}$ in P-NT companion identifications of hatted weight-5 multiplets). Discriminating between these two options is the next concrete arithmetic test of v7.5.

9 Conditional analytical τ derivations: two H_6-co-dependent partial chains (A54)

Two analytical chains have been explored to derive the $\tau_S = i$ fixed point from arithmetic principles. Both are *partial* and *H_6-co-dependent* – they do not constitute "two independent analytical derivations". The v7.5 amendment records this honestly per A54 scoping.

Chain A: Galois descent on the $\Gamma(4)$ modular curve (P-NT). A bijective map between hatted weight-5 S_4' multiplets and a pair ($f_{4.5.\mathrm{b.a}}$, $f_{16.5.c.a}$) of LMFDB CM newforms over $\mathbb{Q}(i)$ [13] forces $\tau_S = i$ via the unique $\Gamma_0(4)$-fixed point of the Galois action; the χ_4 nebentypus restriction H_6 selects $\alpha_2 = 1/12 = B_2/2$ as the canonical algebraic part. This is the chain written up in the companion P-NT paper (BLMS submission queue).

Chain B: Bost–Connes $\times$ CM at $\beta = 2\pi$ (A54 scoping). A new chain uses the Connes–Marcolli–Ramachandran KMS-states-$\times$-complex-multiplication framework [34] (`arXiv:math/0501424`) for imaginary quadratic K including $K = \mathbb{Q}(i)$. The five-step chain is: (S1) BC$\times$CM construction admits a $\beta = 2\pi$ formulation (✓ via CMR05); (S2) the 2-fold extremal-KMS ambiguity i vs. $-i$ for $h(\mathbb{Q}(i)) = 1$ (open); (S3, dominant obstruction) $\beta = 2\pi$ does NOT pick out infinity-type $(4, 0)$; the H_6 χ_4 nebentypus is still required as input – Chain B does *not* improve on Chain A's H_6-dependency; (S4) the type-$\mathrm{III}_1/\mathrm{II}_\infty$ obstruction reappears when identifying $\beta = 2\pi$ between the ECI-side DEHK [25] type-II_∞ algebra and the CMR05 type-III_1 algebra of BC$\times$CM (the bypass of A43 [26] is only PARTIAL); (S5) the Grössencharakter $\psi \to f_{4.5.\mathrm{b.a}}$ θ-lift via Hecke 1937 (✓).

Honest verdict. Chain B is genuinely new and publishable as a conditional math-ph note in 6–8 weeks sub-agent budget; full closure as an analytical theorem requires 24+ months and a co-author (Marcolli, Neshveyev). The corrected probability of closure as a clean derivation of $\tau = i$ *independent of* H_6 is $\sim 12\%$, down from the earlier Opus scoping estimate of 25%. v7.5 therefore records the chain as a SECOND H_6-co-dependent partial derivation, not a second independent one.

Brief-internal correction (logged not propagated). The earlier internal scoping had cited `arXiv:hep-th/9407124` as "Bost–Connes 1995" – this ID is in fact Gelfand–Krob–Lascoux–Leclerc–Retakh–Thibon "Noncommutative symmetric functions". The actual Bost–Connes 1995 paper (Selecta Math. NS 1.3, 411–457) is pre-arXiv. The chain's primary reference is CMR05 (`arXiv:math/0501424`), which generalises BC$\times$CM to imaginary quadratic K including $\mathbb{Q}(i)$. No fabricated bibliography entry was produced; the brief-internal mis-tag is logged in A54 SUMMARY (hallu counter unaffected).

10 Adelic Katz p-adic L-function and Conjecture M13.1 (v7.6, paper-2 candidate)

The Phase 3.B Wave 12 Opus deep-dive (M13, 2026-05-06) and the subsequent audits (M20–M22) identified a sharp 2-adic mathematical problem on $f_{4.5.\mathrm{b.a}}$ with a precise rescue conjecture.

This section records the conjecture for the math.NT paper-2 (target venue: *Algebra & Number Theory*, ANT-tier).

The Frobenius zero-root degeneracy at $p = 2$. The local Euler factor of $f_{4.5.b.a}$ at $p = 2$ has characteristic polynomial $X^2 - a_2 X + \chi_4(2)\, 2^{k-1}$. For $f_{4.5.b.a}$ at $p = 2$: $a_2 = -4$, $\chi_4(2) = \mathrm{Kronecker}(-4, 2) = 0$ (because $2 \mid \mathrm{cond}(\chi_4) = 4$), so

$$X^2 + 4X + 0 = X\,(X + 4),$$

with one root identically zero (NOT a numerical accident; an arithmetic consequence of $p = 2 \mid N$ and $p = 2$ ramified in $K = \mathbb{Q}(i)$). The classical Katz 1978 construction of $L_p(f)$ uses the *unit root* of Frobenius; here, no unit root exists. Standard extensions (Pollack 2003; Sprung 2015; Lei–Loeffler–Zerbes 2010 `arXiv:0912.1263`; Andreatta–Iovita 2024 `arXiv:1905.00792`; Pollack–Rubin 2004 `math/0509604`; Burungale–Büyükboduk–Lei `arXiv:2310.06813`; Büyükboduk–Neamti 2026 `arXiv:2604.13854`) either explicitly exclude $p = 2$, or require $p \nmid N$, or require $a_p = 0$. The triple $\{k = 5,\ p = 2 \mid N,\ a_p \neq 0\}$ is not covered by any single published framework.

Conjecture M13.1 (Pollack-rescue via Kriz 2021 Hodge filtration). Let $f = f_{4.5.b.a}$ and $\Gamma = \mathrm{Gal}(K_\infty^{\mathrm{anti}}/K) \cong \mathbb{Z}_2$ the anti-cyclotomic $\mathbb{Z}_2$-extension of $K = \mathbb{Q}(i)$. Conjecturally, there exist two 2-adic distributions $L_2^+(f),\ L_2^-(f) \in D(\Gamma, \mathbb{Z}_2)$ such that:

(a) *Interpolation along functional-equation symmetrisation* (M21 finding): $\int_\Gamma \chi_m \, dL_2^\pm(f) = E_2^\pm(f, m) \cdot \frac{\alpha_m \pm \alpha_{k-m}}{2} \cdot \Omega_2^{2m}$ for $m \in \{1, \ldots, k-1\}$, where χ_m is a Hecke character of infinity type $(m, k-1-m)$, $E_2^\pm(f, m)$ is a new Euler-type factor compensating for the $X(X+4)$ zero-root degeneracy, and $\Omega_2 \in \mathbb{Q}_2^\times$ is a 2-adic period from the formal group of the elliptic curve $E : y^2 = x^3 + x$ at $p = 2$.

(b) *Boundedness via Iwasawa-log convention* [TBD: prove]: $\log_p(\alpha) = \log_p(-4) = 0$ in the Iwasawa convention; use $\log_p(\gamma)/(\gamma - 1)$ as renormalising denominator (analogue of Pollack's $\omega_n^\pm$ construction).

(c) *Damerell ladder consistency post-renormalisation* (M22 finding): for the F_1 renormalisation $\alpha_m^{\mathrm{ren}} = \alpha_m \cdot (-2^{m-1}) \cdot (1 + 2^{m-3})$, the 2-adic valuations follow the strictly monotone $v_2(\alpha_m^{\mathrm{ren}}) = \{-3, -2, 0, +1\}$.

(d) $N = p^2$ *structural prerequisite*: the construction is well-defined precisely because $f_{4.5.b.a}$ is the unique CM-by-$\mathbb{Q}(i)$ weight-5 newform with $N = p^2 = 4$ in the small-level regime (§3.4); for non-square N the α_m^{ren} are irrational and (c) is undefined.

Best-framework match (Kriz 2021 + Fan–Wan 2023). The most promising existing framework is Kriz's 2021 monograph *Supersingular p-adic L-functions, Maass–Shimura Operators and Waldspurger Formulas* (Princeton AMS-212, ISBN 9780691216478), which constructs p-adic L-functions via Hodge filtration and de Rham/Hodge–Tate periods (NOT Frobenius unit-root) for imaginary-quadratic fields where p is inert OR ramified. Combined with Fan–Wan 2023 (`arXiv:2304.09806`) for the $p = 2$ ramified local-type analysis (the local component of $f_{4.5.b.a}$ at $p = 2$ is a ramified principal series), Kriz's framework bypasses the $X(X + 4)$ zero-root degeneracy entirely. Conjecture M13.1 is a precise, computationally-checkable target for this hybrid framework. [TBD: prove Kriz extends to $p \mid N$.] [TBD: prove Fan–Wan principal-series condition exactly satisfied.]

Paper-2 plan. Title (revised post-M20): *The $N = p^2$ interlinking and a Pollack-type rescue conjecture for the CM newform 4.5.b.a.* Length: 18–22 pp. Target venue: *Algebra & Number Theory* (ANT) primary; backup *Compositio Math.*, *J. Number Theory*, *Res. Number Theory.* Collaborator targeting: Daniel Kriz (MIT) PRIMARY; Antonio Lei (Ottawa), David Loeffler (Sutton), Francesc Castella (UCSB), Kâzım Büyükboduk (UCD) secondary; Joël Bellaïche (Brandeis) for θ-critical interpretation. Outreach scheduled Q3 2026 post-arXiv submission of the present P-NT companion paper (in preparation, math.NT companion).

11 Falsifiers: seven experimental disambiguators 2027–2045

The v7.5 amendment is falsifiable on seven near-future experimental timescales, listed in chronological order with target year, central-value prediction, and confidence-level threshold for refutation.

1. **NA62 $K \to \pi\nu\bar{\nu}$ + FLAG-2027 lattice f_K/f_π on $|V_{us}|$ (2027–2028).** *[Downgraded from "v7.5 prediction" to "speculative observation" per A62+A72.]* The numerical coincidence $|V_{us}| = 9/40 = 0.22500$ (Section 8) is retained as a **speculative numerical observation** ($P_{\text{null}} > 30\%$, A72; full-CKM null $< 1.7\sigma$ vs random, A62) but is **NOT** an ECI structural prediction. NA62 + FLAG-2027 tested at $\sim 0.3\%$ precision by 2028: if $|V_{us}|$ falls outside 0.225 ± 0.001 at 3σ, the coincidence is ruled out, but this does not constitute a falsification of ECI's modular-flavour core (only a numerological accident would be removed). Survival margin ~ 0.5–1.5σ.

2. **KamLAND-Zen final $\langle m_{\beta\beta}\rangle$ + nEXO 2030+ (2027–2030).** v7.5 prediction $m_{\beta\beta} \sim$ few meV (CSD$(1+\sqrt{6})$ at NO with $m_1 = 0$, Section 5.3). Current KamLAND-Zen upper bound $\langle m_{\beta\beta}\rangle < 28$–$122$ meV [35]. A measured $\langle m_{\beta\beta}\rangle > 50$ meV at 3σ from KamLAND-Zen 2027 / nEXO 2030+ refutes the NO + $m_1 = 0$ component of A14 $\Rightarrow$ **REFUTED** (CSD$(1+\sqrt{6})$ row).

3. **JUNO 2030 NMO at 3–4σ (2030).** v7.5 prediction normal ordering with $m_1 = 0$ (Section 5.3). Inverted ordering at $\geq 3\sigma$ by 2030 [37] $\Rightarrow$ **REFUTED** (CSD$(1+\sqrt{6})$ row).

4. **DUNE 2030+ δ_{CP} (2030–2032).** v7.5 prediction $\delta_{CP} \approx -87°$ (Section 5.3; CSD$(1+\sqrt{6})$). DUNE 2030+ at $\pm 15°$ [23] on δ_{CP} rules out the prediction at $> 3\sigma$ if δ_{CP} centres at 0 or $+90° \Rightarrow$ **REFUTED** (CSD$(1+\sqrt{6})$ row). Sensitivity sufficient by ~ 2032. The leptogenesis $(n-1)^2 = 6$ signature (Section 5.6) inherits this falsifier indirectly through the joint $(\theta_{12,13,23}, \delta_{CP}, m_2, m_3, Y_B)$ global fit: if the seven-observable χ^2/dof prefers $n = 1+\sqrt{6}$ over $n = 3$ at $> 2\sigma$ by 2030+, A14 + A55 are corroborated; if χ^2/dof degrades by $> 3\sigma$ relative to $n = 3$, $\Rightarrow$ **REFUTED**.

5. **CMB-S4 Σm_ν at ~ 0.04 eV (~ 2032).** v7.5 prediction $\Sigma m_\nu \approx 0.06\,\text{eV}$ (CSD$(1+\sqrt{6})$ NO floor, Section 5.3). CMB-S4 [36] 2032+ at $\sigma(\Sigma m_\nu) \sim 0.04$ eV discriminates NO + $m_1 = 0$ (preferred prediction 0.058 eV) from inverted with $m_3 = 0$ (0.10 eV) at $\sim 2\sigma$. A measured $\Sigma m_\nu > 0.10$ eV at $3\sigma \Rightarrow$ **REFUTED** (CSD$(1+\sqrt{6})$ row).

6. **LHCb-U2 / Belle II run 3 on $|V_{cb}|^2$ (2030+).** *[A62 RETIRED: $|V_{cb}|^2 = 1/600$ is no longer an ECI v7.5 prediction.]* The v7.4 entry $|V_{cb}|^2 = 1/600$ is RETIRED per A62: HFLAV-2025 average $|V_{cb}| = 0.04183(79)$ places 1/600 at 1.26σ (was $< 0.1\sigma$ on HFLAV-2024). The improved candidate $|V_{cb}|^2 = (7/4)\alpha_1^3 = 7/4000 = 0.00175$ at 0.0038σ (A62) is noted for record but *not* promoted to prediction (null test, A62+A72). This falsifier row is **VOID** for v7.5; it is kept for audit continuity with v7.4 only. Belle II 2027 at $\sim 0.5\%$ and LHCb-U2 / Belle II run 3 (2030+) at $\sim 0.2\%$ precision will constrain $|V_{cb}|^2$ but are not ECI discriminators at this stage.

7. **Hyper-K + DUNE proton-decay branching ratio (2030–2045).** v7.5 prediction (post-G1.12.B M_6) $B(p \to e^+\pi^0)/B(p \to K^+\bar{\nu}) \in [0.3, 3]$ at 95% CL (Section 5.2). Outside $[0.3, 3]$ at $> 3\sigma$ by Hyper-K + DUNE combined 2030+ data, OR $\tau(p \to e^+\pi^0) > 3\times10^{35}$ yr at $> 2\sigma \Rightarrow$ **REFUTED**. Decisive Hyper-K 20-yr measurement $\sim$ 2045. This is the sole Nobel/Foundational-class testable prediction in the portfolio; G1.12.B M3–M6 closure (3.5–5 sub-agent months) is the immediate work-front.

To these we append a continuing v7.4 falsifier:

(8) NMC cosmological NULL at Cassini. A future improvement of the Cassini bound to $|\gamma - 1| \leq 10^{-6}$ that excludes the metric-NMC NULL falsifies the cosmological branch of v7.5-metric (but NOT v7.5-P; see Section 13).

12 A72 honest negative result: Damerell-ladder numerology indistinguishable from random on fermion observables

This section records a *clean, definitive negative result* (A72, 2026-05-06) following the scientific norm of pre-emptive disclosure. Negative results of this type strengthen, rather than undermine, a research programme by eliminating unpromising directions and sharpening the surviving claims.

Test setup. The A72 null test constructed the full Damerell-derivable invariant space for $K = \mathbb{Q}(i)$: 150 candidates built from all products $\alpha_i \cdot \alpha_j$, $\alpha_m \cdot \pi^k$, $\alpha_m \cdot \sqrt{n}$ (for $n \leq 10$), $\alpha_m \cdot |L_p|$, and w_K Weierstrass factors. These were compared against 19 low-energy fermion observables: PMNS angles $(\theta_{12}, \theta_{13}, \theta_{23})$, Dirac phase δ_{CP}, Jarlskog J_{PMNS}, sum Σm_ν, ratio $\Delta m^2_{21}/\Delta m^2_{31}$, charged-lepton ratios m_μ/m_e, m_τ/m_μ, CKM Jarlskog J_{CKM}, and quark Yukawa ratios m_u/m_d, m_s/m_d, m_c/m_s, m_b/m_c, m_t/m_b, y_u/y_t, y_c/y_t, y_d/y_b, y_s/y_b (PDG 2024, NuFit-5.3, HFLAV 2025). For each invariant–observable pair, a rational quotient $q = \text{inv}/\text{obs}$ is computed; a hit is counted if $|\sigma_{\text{hit}}| < \text{threshold}$ and $|q| \leq q_{\max}$.

Headline numbers.

Cut	Threshold	$Q(i)$ hits	Null mean ± sd	σ vs mean
Lax	$\sigma < 0.5$, $\lvert q\rvert \leq 5/4$	718	698.5 ± 42.9	+0.46
Strict	$\sigma < 0.3$, $\lvert q\rvert \leq 1$	**358**	366.2 ± 25.0	**−0.33**

Null distribution: 512 Monte Carlo trials replacing the $K = \mathbb{Q}(i)$ Damerell ladder with random small-rational ladders of comparable magnitude (seed 20260505, deterministic). Strict-cut tail probability $P_{\text{null}} = 0.63$: $Q(i)$ is *below* random mean and wholly consistent with chance. Full posterior in `A72_DAMERELL_EXTENSION/null_test_results.json`.

Verdict. STRONG DEAD-END. The $K = \mathbb{Q}(i)$ Damerell-ladder numerology is statistically indistinguishable from random rational ladders on the full PMNS+lepton+quark observable space. This is a stronger result than A62 (1.7σ on 2 CKM observables) because it covers 19 observables and 150 invariants with a 512-trial null distribution.

What this kills and what it preserves.

- **Killed:** any claim that the Damerell ladder $\{\alpha_m\}$ *predicts* or *structurally anchors* low-energy fermion observables ($|V_{us}| = 9/40$, $|V_{cb}|^2 = 1/600$, PMNS-sector α_m claims). The H_7' axiom status is downgraded from structural input to notation convention (Section 8).

- **Preserved:** the LMFDB 4.5.b.a CM newform anchor, Hecke H_1 restriction (χ_4 nebentypus), and the Cardy hits $\alpha_2 = 1/12$, $\alpha_3 = 1/24$ as *mathematical* identities verified at dps = 60. These are arithmetic anchors, not predictions. The remaining six falsifiers (items 2–5, 7, 8 of Section 11) are unaffected.

13 Outlook and open questions

We collect the open questions and conditional research directions that remain after v7.5.

Q1: dMVP26 quark CP-phase compensator (Section 6). Quantitatively, does a G1.12.B M3–M4 RGE matching reproducibly displace the modulus to $\mathrm{Re}\,\tau \sim 10^{-3}$ at the GUT scale, recovering the observed Jarlskog $J^q_{CP} \sim 3 \times 10^{-5}$ via the linear scaling of [5] p. 9? Alternatively, does the lepton-sector CP-spurion route (a) feed the CKM phase via a GUT-scale Yukawa-mixing term? This is a 12–16 week sub-agent campaign post-G1.12.B M3–M4 closure.

Q2: BDP vs. second-form discrimination of $c \in \{7/10, 4/5\}$ (Section 8). Either the BDP p-adic anti-cyclotomic complement supplies these unfit central charges via Iwasawa-algebra interpolation, or the v7.5 anchor must extend to a two-form anchor $(f_{4.5.\mathrm{b.a}}, f_{16.5.c.a})$. Discriminating (a) from (b) is the next concrete arithmetic test.

Q3: A54 BC×CM scoping memo (Section 9). 4-week Sonnet sub-agent to formalise the conditional math-ph note (publishable in 6–8 weeks if the chain is at least *conditionally* closed). Q1 2027 target.

Q4: v7.5-P Palatini sub-branch decision matrix (Section 2). The Palatini sub-branch makes four new predictions (P1–P4) under the assumption that the Wolf et al. 2025 large-ξ signal [24] is real:

- P1: under a Palatini posterior, $\xi_\chi \in [+1.5, +3.0]$, $\log B > +5$ vs. ΛCDM.
- P2: $G_{\mathrm{eff}}(z = 0)/G_N \in [1.4, 1.8]$ in DESI Y3 + LSST 2030.
- P3: $|\gamma - 1| < 10^{-9}$ at 1 AU (Palatini Yukawa screening).
- P4: all v7.5 arithmetic-modular falsifiers (DUNE δ_{CP}, Belle II V_{cb}, Hyper-K p-decay, CSD$(1 + \sqrt{6})$, Damerell ladder) preserved unchanged.

Decision matrix:

- Wolf signal independently confirmed by 2027 (e.g. ACT DR6 + DESI Y3) $\Rightarrow$ COMMIT v7.5-P as primary cosmological branch.
- Wolf signal refuted (e.g. ACT DR6 finds no 4σ enhancement) $\Rightarrow$ DROP v7.5-P; v7.5-metric remains baseline.
- All v7.5 arithmetic predictions close NEGATIVE by 2030 $\Rightarrow$ DROP entire programme; v7.5-P moot.

Q5: Leptogenesis sharper falsifier (Section 5.6). The $(n-1)^2 = 6$ signature is indirect; a sharper falsifier requires either a derivation of the high-energy phase η from ECI primitives (none known), or a cosmological probe of leptogenesis distinct from BBN Y_B. 4-6 week sub-agent campaign A56 in queue (NMC retrain Wolf-large-ξ + leptogenesis joint).

Q6: Modular-Higgs assignment as Axis-1 extension. The Higgs doublet H as an S_4' non-trivial representation, in the sense of Penedo–Petcov 2020 [3] or Kobayashi–Tanimoto, would constrain λ_H modular-form-wise. Cost $\sim$ 8–12 months sub-agent + Higgs-sector modular-form expert (Penedo, Petcov, King, Tanimoto). Foundational-Prize relevance LOW unless the prediction is at $< 5\%$ precision; current PDG M_W/M_Z precision is 0.01%, so any modular prediction is unlikely to be falsifiable as new physics. We register this as a deferred Q3 2026 outreach item.

Q5: SUSY commitment (v7.6 NEW, post-O1 D2). The v7.6 amendment commits to **NO-GO** on full MSSM supersymmetry, with split-SUSY conditional fallback if Hyper-K 2045+ confirms $\tau(p \to e^+\pi^0) \approx 6 \times 10^{34}$ yr *and* a precision $\alpha_s(M_Z)$ measurement at 0.1% fails to close 2-loop unification. Quantitative defence (per O1 D2.3): ECI's modular sector at $\tau \approx i$, the A14 CSD$(1+\sqrt{6})$ Littlest Modular Seesaw at the high-scale, and the A73 RG-stable wedge $\xi_\chi \in [-0.029, +0.001]$ together provide the structural protections that low-energy MSSM provides in conventional SU(5) GUTs: (i) discrete-symmetry protection of small Yukawa from radiative corrections via S_4' weight assignments; (ii) high-scale neutrino mass without low-scale superpartners; (iii) RG-stable non-minimal coupling without sparticle threshold corrections. The price is gauge unification at 2-loop precision rather than 3-loop, plus a tuned doublet-triplet split — both inherited from non-SUSY minimal SU(5). Adding MSSM would triple the parameter count (+5 mSUGRA $\to$ +20–30 post-LHC) and undermine the modular-flavour minimality narrative; the LHC Run 3 null-result on light sparticles makes vanilla MSSM unattractive on its own. Migration to split-SUSY (Arkani-Hamed–Dimopoulos `hep-th/0405159`, Giudice–Romanino `hep-ph/0409232`) preserves modular minimality at the cost of three additional high-scale parameters ($\tilde{m}, \mu, \tan\beta, A$-term) and is pre-registered as the conditional 2030+ fallback.

Q6: Honest tensions audit (v7.6 NEW, post-M2 falsifier matrix). Two non-falsifying tensions are explicitly recorded:

- **Wolf25 ξ tension (5 OOM, cosmological, non-falsifying)**: Wolf 2025 (`arXiv:2504.07679`) reports $\xi = 2.30^{+0.71}_{-0.38}$ on DESI DR2 + CMB + DES-Y5; ECI metric-branch $\xi \sim -3 \times 10^{-5}$ is $\sim$ 5 OOM outside this posterior. However, the M7 + M9 Wolf-NMC-KG ODE diagnostic demonstrates Wolf's $\xi = 2.31$ is *KG-failing* (Klein-Gordon ghost $F < 0.01$ within the prior) whereas ECI's $\xi \sim 0.001$ is *KG-passing*. Resolution path: NMC Bayes contest, queued in v7.7. ECI *and* the Antusch-King-Wang 2026 coupled-DM-DE model (`arXiv:2604.08449`) are both in the KG-passing class; Wolf is structurally disqualified.

- **Σm_ν near-tension ($\sim 1.7\sigma$, non-falsifying)**: ECI A14 predicts $\Sigma m_\nu = 65$–69 meV; DESI DR2 + CMB 95% UL is 64.2 meV in ΛCDM. ECI's lower band is ~ 1 meV above the LCDM UL — formally a $\sim 1.7\sigma$ tension. In the $w_0 w_a$CDM background the UL relaxes to ~ 160 meV (fully consistent). CMB-S4 (2032+) projects $\sigma(\Sigma m_\nu) \sim 15$ meV: ECI becomes a $\sim 4\sigma$ *positive prediction* test by 2035 if normal ordering is confirmed.

Closing remark. The v7.6 amendment consolidates the v7.4–7.5 architecture, retracts the LYD20 unified-scaffold graft per A46, integrates the Phase 3 deliverables of 2026-05-06 (M7 Wolf-NMC-KG ODE, M9 three-way reframe, M11 A55 grid 1024, M13 OPUS deep adelic, M17 Modular Shadow v2.5, M21 functional-eq symmetrisation, M22 $N = p^2$ META-finding, M23 B-ratio audit, M24 ξ_χ rail diagnostic), commits to SUSY NO-GO with split-SUSY conditional fallback, and presents the first real-data ECI posterior with $H_0 = 68.51 \pm 0.14$ and a ξ_χ rail-saturated at the Cassini boundary interpreted via A73 RG-running. The v7.4 modular-flavour body and Karam-Palatini + KW + dMVP26 + A55 grafts carry over. A post-submission negative result (A72, Section 12) establishes that the $K = \mathbb{Q}(i)$ Damerell-ladder numerology does not exceed random on the full PMNS+lepton+quark observable space ($\sigma = -0.33$, $P = 0.63$): the H_7' axiom is downgraded to notation convention and the A17 CKM alignments are

demoted to speculative observations. The surviving arithmetic core ($f_{4.5.b.a}$ LMFDB 4.5.b.a, Hecke H_1, χ_4 nebentypus structure, Cardy identities $\alpha_2 = 1/12$, $\alpha_3 = 1/24$) is unaffected. The immediate work-front is G1.12.B M3–M6 closure (3.5–5 sub-agent months) toward the proton-decay branching ratio prediction at Hyper-K 2045+.

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Data and code availability

Repository: `https://github.com/AIdevsmartdata/crossed-cosmos`. Snapshot of v7.4–7.5 axioms: Zenodo DOI `10.5281/zenodo.20030684` (commit 6.0.53.7, 2026-05-05). v7.5-specific evidence: A46 `A46_LYD20_GRAFT/lyd20_pinned_results.json` (99,061× penalty); A47 `A47_KW_DSTRAP_COMPAT/` (KW `arXiv:2310.10369` live-fetch); A48 `A48_DMVP26_KAHLER_TAU_I/` (`kahler_tau_i.py`, `diag.py`); A50 `A50_KARAM_PALATINI/` (`section2_patch.tex`, `v75P_sketch.md`); A54 `A54_BC_CM_BETA_2PI/SUMMARY.md` (CMR05 `arXiv:math/0501424` chain analysis); A55 `A55_LEPTOGENESIS/` (`lepto_eta_B.py`, `v75_section_lepto` `lepto_summary.json`); A36 `A36_G112B_M5/` (`proton_decay_widths.py`, `tau_p_results.json`); A62 `A62_CKM_SYSTEMATIC/` (`null_test.py`, `ckm_hits.json`, `ckm_headline.json` — CKM-Damerell null 1.7σ; $|V_{cb}|^2 = 1/600$ RETIRED); A72 `A72_DAMERELL_EXTENSION/` (`null_test_optimized.py`, `null_test_results.json` — full PMNS+Q+L null, $\sigma = -0.33$, $P = 0.63$, 512 trials, 0.8s). v7.4 baseline: cf. v7.4 amendment v2 [1].

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